He–vacancy interactions in pure vanadium and tantalum

1 Introduction

Vanadium and its alloys are promising candidates as blanket for fusion reactor thanks to their mechanical properties at high temperature and low activation [26]. Under irradiation, they are subject to damages such as swelling [4], embrittlement [22] and blistering [14] triggered, enhanced or suppressed [6] by He resulting of $(n,\ \alpha)$ transmutation reactions. He atoms are usually trapped at grain boundaries, dislocations or void clusters [1, 9]. Experiments are unable to resolve the early stages of formation of He bubbles in metals, but *ab inito* calculations are powerful tools to investigate the processes taking place at the beginning of nucleation of such objects. Only in recent years, such studies on vanadium began [12, 41, 42, 44] and there is few literature concerning tantalum. In this study we will focus on He-vacancy type clusters in pure vanadium and tantalum, in particular we are interested in the first steps of nascent bubbles.

This paper is organised as follows : Sec.2 gives the parameters of the methodology employed for the calculations. In Sec.3.1 we present the formation energies computed for vacancy and He-vacancy type clusters. In Sec.3.2 the stability of He-vacancy type clusters is investigated. Finally, Sec.3.3 describes the migration processes and barriers of vacancy clusters and how helium modifies the diffusion mechanisms.

2 Computational details

Ab initio calculations were performed with density functional theory (DFT) as implemented in the Vienna *Ab initio* Simulation Package (VASP) [16, 17]. We employed non spin-polarised calculations with the generalised gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) [27] exchange-correlation functional, and semi-core projector-augmented wave (PAW) [2, 18] as pseudo-potentials. According to Ref.[43], Van der Waals interactions (~ 20 meV) can be neglected. For simulation modelling up to 3 vacancies, a bcc supercell of 128 atoms with a $4 \times 4 \times 4$ k-points mesh grid was used; for solid with more than 3 vacancies we used a 250 atoms supercell with a $3 \times 3 \times 3$ k-points mesh grid generated by Monkhorst–Pack scheme [24]. The Methfessel-Paxton smearing [23] with a 0.25 eV width was used and a 350 eV cutoff energy for both V and Ta case. Atomic position were relaxed at constant volume until forces on each atom were less than 0.01 eV. The migration barriers and paths were determined using the climbing image nudged elastic band method (CI–NEB) [10, 11]. In addition, the converged energy was corrected by subtracting the elastic dipole–dipole interaction energy between self-images due to periodic boundary conditions [38]. The equilibrium lattice constant obtained for V was 3.00 Å and 3.32 Å for Ta are in agreement with experimental data (respectively 3.03 Å and 3.30 Å) [15] .

3 Results

3.1 Formation of vacancies and Helium clusters

The formation energy of He at high symmetry interstitial sites in bcc lattice is given by :

$$E^f = E_{He_n} - E_0 - n \times E_{He} \tag{1}$$

where n denotes the number of He atom, E_{He_n} energy of the supercell with n interstitials cluster, E_0 energy of the perfect supercell and E_{He} energy of an isolated He atom. The formation energies are respectively in V E^f=3.12 eV for tetrahedral site (T) and E^f=3.34 eV octahedral site (O), in Ta E^f=3.47 eV (T) and E^f=3.79 eV (O). As for other bcc transition metals such as Fe [8], W and Nb [30], the T site is more stable at low temperature than the O site for both V and Ta.

We also studied the formation of vacancy clusters (V_m) in both V and Ta, up to m=4 defined by :

$$E^f = E_{V_m} - \frac{(N-m)}{N} E_0 \tag{2}$$

with N the number of atoms in the perfect lattice and E_{V_m} energy of the supercell with m vacancies cluster. For each type of clusters, we take only the most stable one to study the interaction with He in the following. We find out that the lowest energy cluster to be the most compact (see Fig.1), except for the di-vacancy cluster for which the 2nd nearest neighbour cluster (2nn) is lower than the 1st nearest neighbour one (1nn) by 0.19 eV in vanadium and 0.24 eV in tantalum (Tab.1).

We now study multiple He-vacancy clusters (up to n=26 for m=4) for which the formation energy is defined as :

$$E^f_{He_nV_m} = E_{He_nV_m} - \frac{(N-m)}{N} \times E_0 - n \times E_{He} \tag{3}$$

where $E_{He_nV_m}$ is the energy of the supercell with the He_nV_m cluster.

As the number of clusters (He_nV_m) increases very fast with n and m, we picked tens of configurations of each type, selected randomly with He atoms placed at

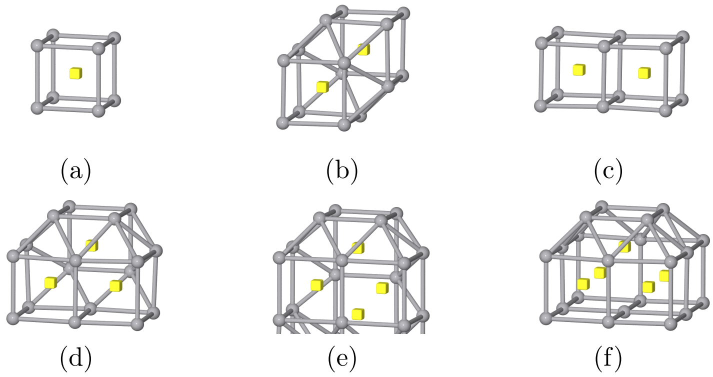


Figure 1: Lowest formation energy vacancy type clusters V_m. (a) V_1, (b) V_2(1nn), (c) V_2(2nn), (d) V_3, (e) V_4, (f) V_5. Vacancies are represented with yellow cubes.

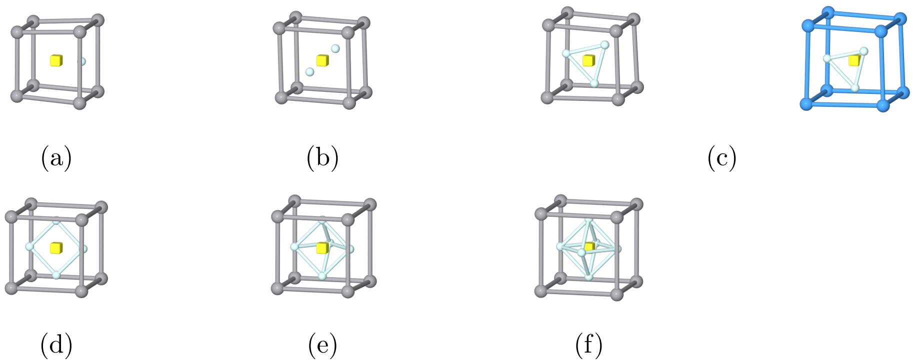


Figure 2: Lowest formation energy He_nV_1 type clusters in vanadium and tantalum. (a) He_1V_1, (b) He_2V_1, (c) He_3V_1, (d) He_4V_1, (e) He_5V_1, (f) He_6V_1. Vacancy is represented with yellow cube, vanadium with grey ball, tantalum with dark blue ball and helium with light blue ball. Configurations for tantalum are shown only when different to vanadium ones.

first nearest neighbour T, O or substitutional (S) sites to the vacancy.
For He_1V_1 in vanadium, the He atom is slightly off O site (Fig.2) in the lowest energy configuration (E^f=4.47 eV) rather than at S site (E^f=4.68 eV) while in tantalum, the substitutional site is the only stable configuration.

For He_2V_1 cluster the 2 He atoms and the vacancy are aligned in the <111> direction. The lowest energy clusters from n=4 to n=6 have all He atoms on O sites. In the case of di-vacancy clusters in vanadium, He_1V_2 in the 1nn configuration is still higher in energy than in the 2nn one (Fig.3) but the difference is reduced to 0.09 eV compared to 0.19 eV in the pure vacancy cluster case.

Furthermore, their formation energy are reversed since n=2 as the 1nn configuration is 0.4 eV lower than the other one (Tab.1).
In tantalum, the hierarchy in the formation energy between 1nn and 2nn configuration is already changed for n=1.

For all types of cluster He_nV_m, the cost in energy to form $He_{n+1}V_m$ increases

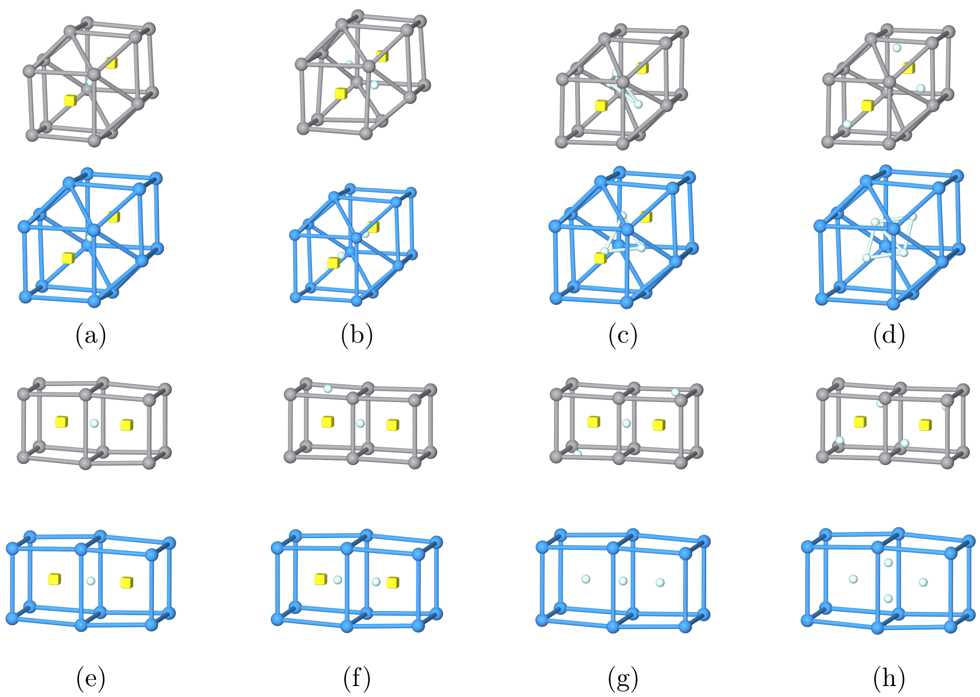
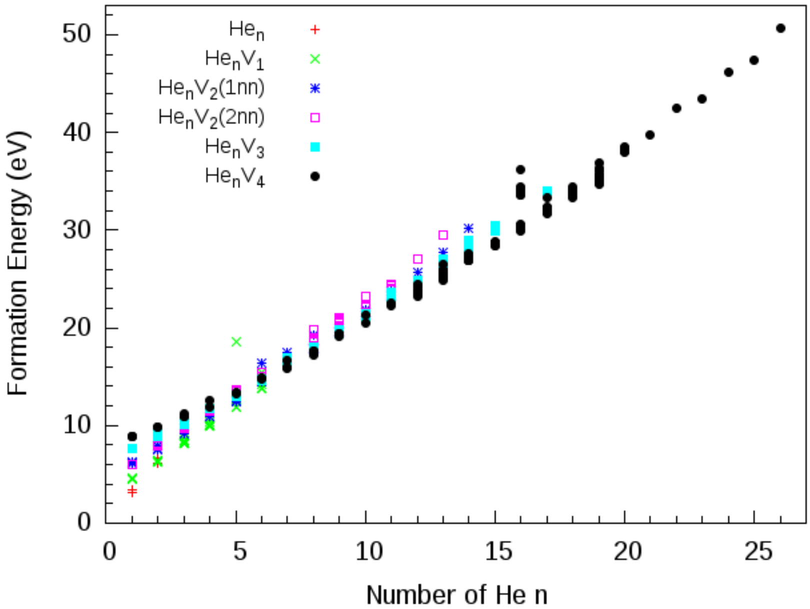


Figure 3: Lowest formation energy He_nV_2 type clusters in vanadium and tantalum. (a) He_1V_2(1nn), (b) He_2V_2(1nn), (c) He_3V_2(1nn), (d) He_4V_2(1nn), (e) He_1V_2(2nn), (f) He_6V_2(2nn), (g) He_3V_2(2nn), (h) He_4V_2(2nn). Vacancy is represented with yellow cube, vanadium with grey ball, tantalum with dark blue ball and helium with light blue ball.

	V					Ta				
n \ m	1	2 (1nn)	2 (2nn)	3	4	1	2 (1nn)	2 (2nn)	3	4
0	2.47 $(2.1\pm0.2)^a$ $(2.2\pm0.4)^b$	4.69	4.50	6.46	8.07	2.88 $(2.8\pm0.6)^a$ $(3.1)^c$ $(2.9\pm0.4)^d$	5.66	5.42	7.97	9.96
1	4.47	6.15	6.06	7.58	8.88	4.66	6.89	6.92	8.77	10.44
2	6.26	7.54	7.94	8.90	9.81	6.49	8.22	8.40	9.97	11.31
3	8.15	9.18	9.74	10.09	10.84	8.63	9.69	10.07	11.09	12.28
4	10.01	10.92	11.48	11.74	11.92	10.79	11.22	11.80	12.29	13.31
5	11.91	12.47	13.45	12.99	13.20		12.84	14.19	13.81	14.46
6	13.73	14.44	15.07	14.62	14.76	14.87	14.80	15.69	15.39	15.62

Table 1: Lowest formation energy E^f in eV of first smallest He-vacancy clusters He_nV_m in vanadium and tantalum.
[a] Experimental data from Ref [21].
[b] Experimental data from Ref [13].
[c] Experimental data from Ref [29].
[d] Experimental data from Ref [28].



m	0	1	2 (1nn)	2 (2nn)	3	4
$\Delta E^f_{n\rightarrow n+1}$ (eV)	2.97	1.86-2.05	1.76-1.82	1.81-1.88	1.63	1.51-1.66

Figure 4: Formation energy as a function of n in vanadium. For di-vacancy clusters the label (1nn) is for 1st vacancy nearest neighbour and (2nn) for 2nd nearest vacancy neighbour. The table gives the slope for each type of cluster.

linearly with n but the slope decreases with m (see Fig.4 and Fig.5).

3.2 Stability of He–vacancy clusters

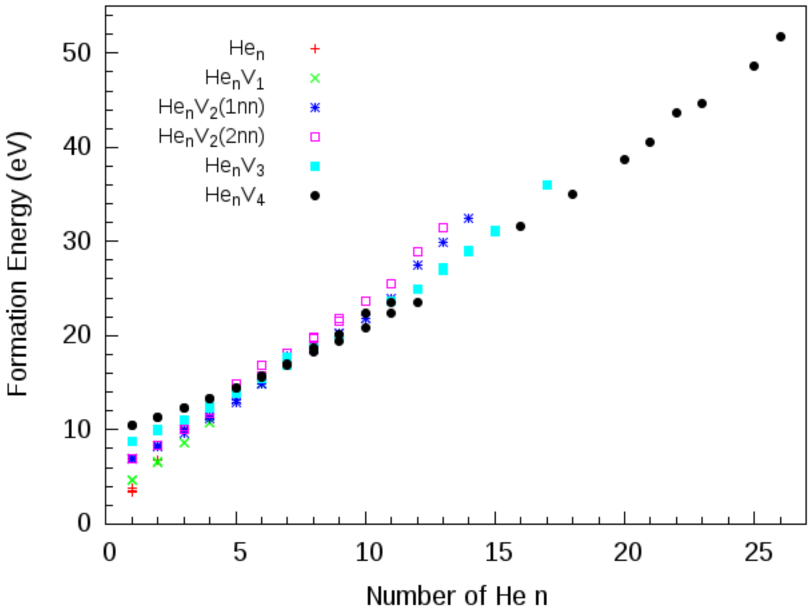
We investigated the stability of pure He and vacancy clusters and He–vacancy complexes looking at their binding energy to helium atom and mono-vacancy given respectively by :

$$E^b_{He-He_nV_m} = E_{He_I} + E_{He_{n-1}V_m} - E_{He_nV_m} - E_0 \tag{4}$$

and

$$E^b_{V-He_nV_m} = E_V + E_{He_nV_{m-1}} - E_{He_nV_m} - E_0 \tag{5}$$

where E_{He_I} is the energy of the supercell with an interstitial He at T site for vanadium and S site for tantalum, E_V is the energy of the supercell with a mono-vacancy. With this definition, a positive binding energy corresponds to an attractive interaction, therefore the clusters with the highest binding energy



m	0	1	2 (1nn)	2 (2nn)	3	4
$\Delta E^f_{n\rightarrow n+1}$ (eV)	3.26	2.06	1.73-2.19	1.87-2.10	1.41-1.90	1.07-1.35

Figure 5: Formation energy as a function of n in tantalum. For di-vacancy clusters the label (1nn) is for 1st vacancy nearest neighbour and (2nn) for 2nd nearest vacancy neighbour. The table gives the slope for each type of cluster.

n \ m	V: 1	V: 2 (1nn)	V: 2 (2nn)	V: 3	V: 4	Ta: 1	Ta: 2 (1nn)	Ta: 2 (2nn)	Ta: 3	Ta: 4
1	1.12	1.64	1.50	2.01	2.32	1.75	2.22	1.95	2.66	2.97
2	1.34	1.73	1.24	1.81	2.21	1.62	2.00	1.98	2.25	2.58
3	1.24	1.49	1.33	1.94	2.10	1.32	2.11	1.77	2.33	2.49
4	1.26	1.39	1.38	1.48	2.04	1.29	1.92	1.72	2.25	2.42
5	1.23	1.58	1.16	1.87	1.85		1.84	1.07	1.93	2.31
6	1.30	1.16	1.51	1.50	1.57		1.49	1.95	1.87	2.29

Table 2: Highest binding energy E^b in eV of first smallest He-vacancy clusters He_nV_m to He in vanadium and tantalum.

are more stable.
A general overview of Fig.6 and Fig.7 shows a broad distribution in He-cluster binding energy for the same type of cluster indicating a strong dependence of the configuration of the clusters. The binding energy of pure He clusters in vanadium range from -0.38 eV to 0.24 eV slowly increasing with the amount of He at least up to n=4, similar to tantalum case, which is 4 times less than in Fe ($\sim$1 eV)[8]. Adding more He amplify lattice distorsion so that the increasing pressure may lead to the creation of Frenkel pair as predicted in Fe [3, 25, 40] and observed in Ni [34] and Au [33].
Considering each type of cluster with the highest binding energy, for each n, the binding energy increases with the number of vacancy (Tab.2). As a global trend, one can see for a given m, the binding energy decreases with n as a consequence of an increase of internal pressure in the cavity and spontaneous emission of He or self-interstitial may occurs for large n/m ratio [31, 32]. This behaviour has been observed at low temperature [7] and also predicted in Fe [25]. Nevertheless, we should highlight that considering the restricted sample taken within the many possible isomers, there is no guarantee we selected the most stable clusters, in particular for the largest clusters.
In the di-vacancy case, we see again that He make the He_nV_2(1nn) clusters more stable than the He_nV_2(2nn) ones when we have the opposite situation without He. For m=1, in vanadium, the binding energy seems to saturate (up to n=6) indicating that more He atoms can be added before internal pressure reach a threshold such that significant drop in the binding energy occurs. Indeed, previous calculations showed mono-vacancy clusters with tens of He atoms still have attractive interaction with He [20].
He_8V_2(1nn) seem particularly stable. In these clusters He atoms occupy high symmetry sites (O,T), when they are more shifted from these positions in the $He_{20}V_4$ clusters that exhibit a negative binding energy, contrary to $He_{19}V_4$ and $He_{21}V_4$ clusters.

The binding energy of pure vacancy clusters to mono-vacancy increases with the number of vacancy from 0.27 eV to 0.88 eV in vanadium (Fig.8). It seems to reach a plateau at m=4 around 0.87 eV. Similar behaviour has been reported in Fe, using molecular dynamics simulation for which the binding energy saturates slightly above 1 eV for $m \gtrsim 20$ [25].
The presence of He atoms in vacancy clusters enhances the binding energy to mono-vacancy, reflecting the fact that He stabilises vacancy type clusters (Tab.3). We find again He_nV_2(1nn) are more stable than He_nV_2(2nn) for n $\geq$

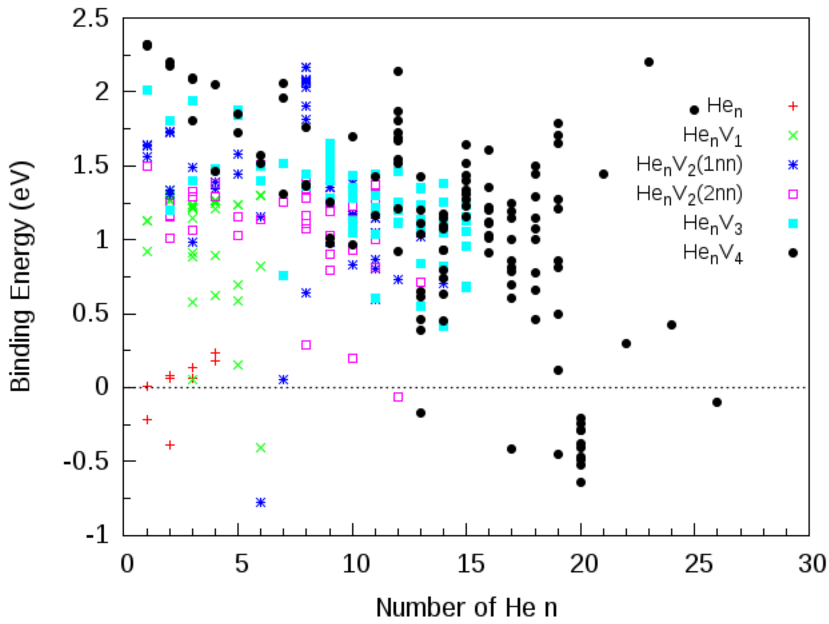


Figure 6: Binding energy of He to He_nV_m clusters as a function of n in vanadium

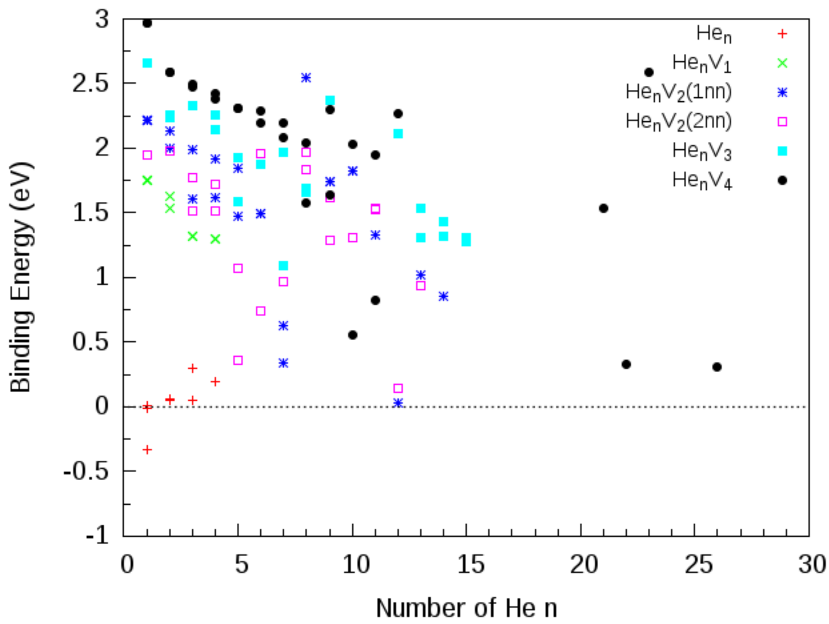


Figure 7: Binding energy of He to He_nV_m clusters as a function of n in tantalum

n \ m	V 1	V 2 (1nn)	V 2 (2nn)	V 3	V 4	Ta 1	Ta 2 (1nn)	Ta 2 (2nn)	Ta 3	Ta 4
1	1.11	0.79	0.89	0.94	1.17	1.74	0.73	0.70	1.11	1.28
2	2.37	1.19	0.79	1.50	1.57	3.31	1.11	1.06	1.39	1.61
3	3.47	1.43	0.88	2.12	1.72	4.33	1.90	1.51	1.94	1.77
4	4.49	1.56	1.00	2.22	2.28	5.43	2.52	1.94	2.47	1.93
5		1.91	0.92	2.93	2.25				3.33	2.32
6		1.76	1.13	2.92	2.32		3.02	2.14	3.26	2.73

Table 3: Highest binding energy E^b in eV of first smallest He-vacancy clusters He_nV_m to mono-vacancy in vanadium and tantalum.

2 in vanadium and n $\geq$ 1 in tantalum.

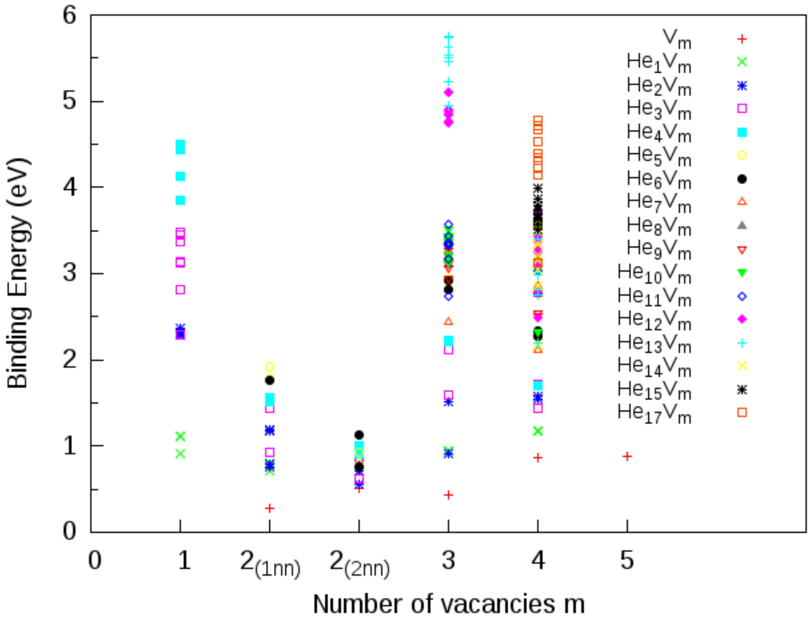


Figure 8: Binding energy of mono-vacancy to He_nV_m clusters as a function of m in vanadium

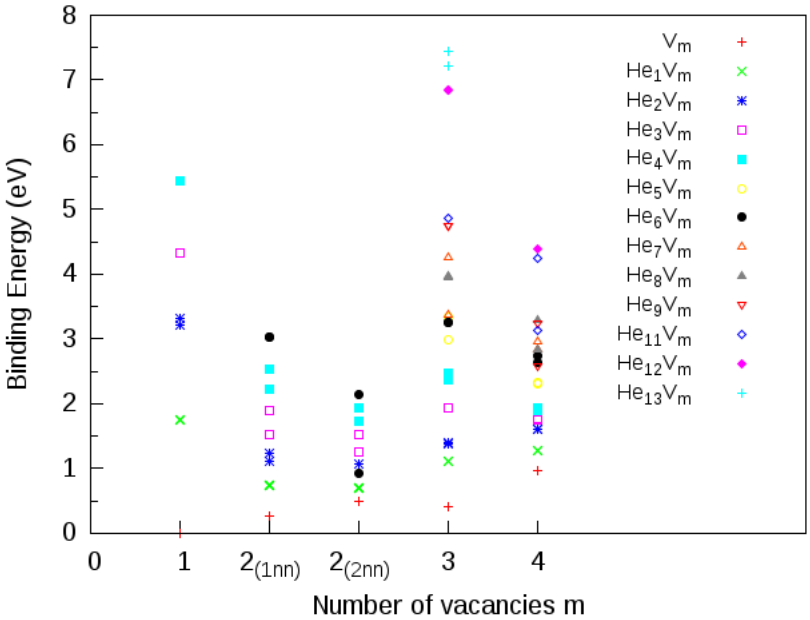


Figure 9: Binding energy of mono-vacancy to He_nV_m clusters as a function of m in tantalum

3.3 Migration energy barriers

It is necessary to capture the diffusion processes of interstitial He and vacancy in order to understand how clusters grow at the primary stages.

Vanadium

We computed 2 paths for interstitial diffusion barrier : T site to T site, and T site to O site. T $\rightarrow$ T path shows a barrier of 0.12 eV (Tab.4) in good agreement with TDS experiment in vanadium ($E^m_{He} = 0.13$ eV) [7], that is twice the barrier in Fe [8], whereas the barrier of T $\rightarrow$ O path is $E^m_{He} = 0.26$ eV. Thus, interstitial He prefers to diffuse via the first neighbour T sites.
Our calculations of mono-vacancy migration barrier ($E^m_V = 0.48$ eV) agrees very well with TDS experiments ($E^m_V = 0.5$ eV) [7] and ($E^m_V = 0.47$ eV) [37].
The di-vacancy migration barrier from the 1nn configuration to 2nn configuration is $E^m_{V_2} = 0.34$ eV. Di-vacancy cluster in 2nn configuration is more mobile since it can migrate without dissociation with a barrier of $E^m_{V_2} = 0.37$ eV.
We turn on the influence of He on vacancy diffusion. The minimum energy path found for the migration of mono-vacancy with helium in their lowest energy configuration at starting and final point; showed first the dissociation of He to the monovacancy. Then He atom diffuse to a position near tetrahedral interstitial, which is followed by the migration of the mono-vacancy to recombine with He. Dissociative mechanism has been also experimentally inferred [19, 39]. The migration barrier is about 1.27 eV that correspond roughly to the He-vacancy binding energy plus migration barrier of He (T $\rightarrow$ T). As a result, He enhances the diffusion barrier of mono-vacancy by 0.67 eV. For the di-vacancy from 1nn to 2nn configuration, there is a 0.79 eV barrier. Migration from 2nn to 2nn configuration is a 2 steps process with a lower barrier $E^m_{V_2} = 0.66$ eV. First, mono-vacancy dissociate from the cluster to reach first nearest neighbour site changing the system into a slightly shifted 1nn configuration. Then, the 2nd mono-vacancy moves to its nearest neighbour meanwhile He atom goes to its final octahedral position between the 2 vacancies. The trapped He atom almost doubles the migration barrier of the di-vacancy. Nevertheless, the presence of He does not change the relative mobility of mono and di-vacancy but moreover, it amplifies the gap.

Tantalum

Same conclusions as vanadium can be made for tantalum for which migration barrier of interstitials are $E^m_{He}(\mathrm{T} \rightarrow \mathrm{T}) = 0.18$ eV and $E^m_{He}(\mathrm{T} \rightarrow \mathrm{O}) = 0.40$ eV (Tab.4). Interstitials He diffuse hopping from T to T sites.
Concerning vacancies, migration barrier is higher for mono-vacancy ($E^m_V = 0.72$ eV) consistent with resistivity annealing experiment ($E^m_V = 0.7$ eV)[5], than for di-vacancy which energy barrier is 0.17 eV lower in 2nn $\rightarrow$ 2nn path.
Diffusion of mono-vacancy with a He atom involve first a dissociation of He to the vacancy. Then He hops via T sites while mono-vacancy jumps to first nearest neighbour site. Finally He recombines with mono-vacancy, following the 'impeded interstitial mechanism' or 'He-vacancy dissociative mechanism' [35, 36]. He enhances the migration barrier of mono-vacancy by 1.17 eV.
The increase of migration energy due to He is 2 times smaller (0.51 eV) for 1nn $\rightarrow$ 2nn path than for mono-vacancy case.

	V			Ta		
Configuration	Path	E^m (eV)	Exp. (eV)	Path	E^m (eV)	Exp. (eV)
Interstitial						
He	T → T	0.12	0.13 [7]	T → T	0.18	
	T → O	0.26		T → O	0.40	
Vacancy						
Mono-vacancy	1nn	0.48	0.5 [7] [36] 0.47 [37]	1nn	0.72	0.7 [5][36]
Di-vacancy	1nn → 2nn	0.34		1nn → 2nn	0.63	
Di-vacancy	2nn → 1nn	0.58		2nn → 1nn	0.87	
Di-vacancy	2nn → 2nn	0.37		2nn → 2nn	0.55	
He-vacancy clusters						
He_1V_1	1nn	1.27		1nn	1.89	
He_1V_2	1nn → 2nn	0.79		1nn → 2nn	1.14	
He_1V_2	2nn → 1nn	0.88		2nn → 1nn	1.12	
He_1V_2	2nn → 2nn	0.66		2nn → 2nn	1.12	

Table 4: Migration energy barrier (E^m) for He, vacancies and He-vacancy cluster in vanadium and tantalum.

Contrary to the vanadium case all paths involving di-vacancy have the same barrier in presence of He, so He_1V_2 clusters can migrate via three ways with the same probability.

Thermal desorption spectra experiment give us direct access to dissociation energy we can defined as the sum of binding energy and migration energy of He atom or vacancy :

$$E^d_{He/V-He_nV_m} = E^b_{He/V-He_nV_m} + E^m_{He/V} \tag{6}$$

On Fig.10 and Fig.11 the dissociation energy of He and vacancy with most stable He_nV_m cluster as a function of n/m ratio is drawn for respectively vanadium and tantalum. This allows us to determine wether a cluster tends to emit a He atom or a vacancy to gain stability, i.e when $E^d_{He} > E^d_V$, the cluster is more likely to dissociate a vacancy. The crossover shows the most stable clusters for ~ 0.9 n/m ratio with $E^d \sim 1.9$ eV in vanadium, close to experimental result $E^d = 1.66$ eV [39], and $E^d \sim 2.5$ eV in tantalum. This is consistent with TDS experiment [7] in which the peaks at the highest temperature were attributed to the clusters with n=m.

4 Discussion

When we compare the behaviour of He in vanadium and tantalum, one can notice S site is attractive in tantalum while it is repulsive in vanadium (Fig.2–3). Similar result has been found [30] and explained by a larger charge density at vacancy for vanadium which energy is close to the Fermi level, thus closed-shell He will have a repulsive interaction. Indeed, He at S site in vanadium

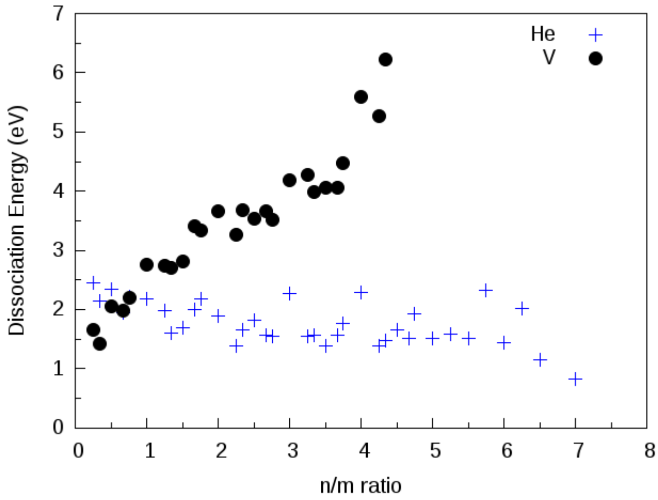


Figure 10: Dissociation energy of He and vacancy to He_nV_m as a function of n/m ratio for the most stable clusters in vanadium.

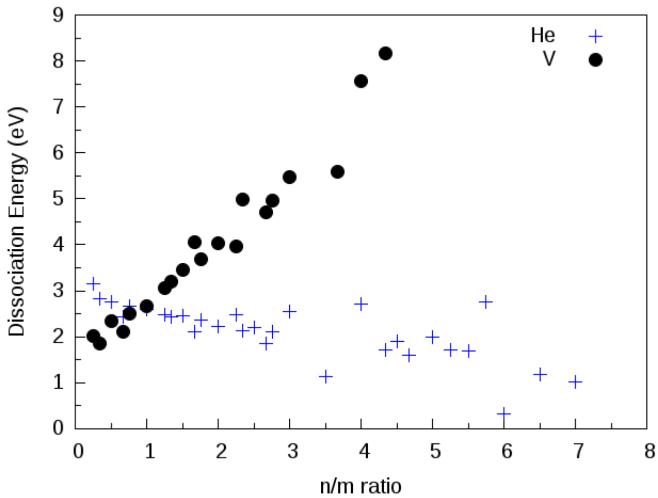


Figure 11: Dissociation energy of He and vacancy to He_nV_m as a function of n/m ratio for the most stable clusters in tantalum.

shows a larger density of state at the Fermi energy level than in tantalum then increasing the formation energy enough to make this configuration unstable. As a result in the most stable configuration, He is shifted toward O site.

5 Conclusion

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